

Examiner reports

Q1.

A surprising number of candidates failed to score both marks for the forces diagram in part (a), with the most common errors being the omission of the friction force or stating the normal reaction force to be equal to the weight. Parts (b) and (c) were done well, and many were successful in part (d), although some omitted a force from their calculations.

Q2.

Diagrams in part (a) were often good, although there was a tendency to include the acceleration of the moving object on force diagrams. Part (b) was generally well done but there was some confusion with signs in part (c). Most completed part (d) but sometimes lost a mark through failing to give their answer to three significant figures as demanded in the rubric for the paper.

Q3.

This question was answered fairly well by the majority of candidates. On the whole the force diagrams were quite good, but a number of classic errors were observed. For example including a force in the direction of motion and putting the normal reaction vertical. The rest of part (a) was answered very well, with the candidates appearing to have been helped by the printed answers.

The answers to part (b) were much more variable. There were quite a few very good answers, which included clear concise explanations.

Q4.

Part (a) was generally well answered, although some candidates did make errors when resolving.

Part (b) was found to be much more difficult. The majority of the candidates did not realise that they needed to include a component of the weight and the friction force acting in the same direction. Also a number of candidates did not seem to realise that the acceleration was directed down the slope.

Q5.

This question was generally done well by the majority of candidates, but some experienced problems. In part (b), there were a few candidates who ignored the instruction to form an equation of motion for each particle. A few others made errors in one equation. Part (c) was usually done correctly if the candidates had two equations of motion in part (b). If they did not have these equations, there were often errors in this part. Part (d) was done well using the acceleration given in the question.

Q6.

Parts (a) and (b) were generally done well. In part (b), the printed answer helped many candidates, but some could not put together a convincing argument. Part (c) was more challenging. Some candidates did resolve horizontally and apply the friction law, but then got into difficulties solving the resulting equation. Other candidates did not take both of the

steps needed to produce an equation for T and made very little progress. There was also some evidence of candidates using $\sin 30^\circ$ instead of $\cos 30^\circ$ when resolving horizontally.

Q7.

The force diagram asked for in this question was drawn fairly well, but the usual errors of forces in the direction of motion and missing forces were present on some scripts. Parts (b) and (c) were done fairly well and it appeared that many 'candidates' had been helped by the printed answer in part (b). There were some candidates who did not realise that they should consider the tension in the rope when finding the normal reaction. Part (d) was found to be very much more demanding, with relatively few correct responses.

Candidate typically omitted one term from their equation or failed to solve correctly for T .

Q8.

The candidates found this question a little harder than the previous two questions. There were a number of poor force diagrams, some showing a force in the direction of motion. The errors made by candidates were mainly due to incorrect resolving, but some of these were corrected to produce the answers given in the question. A small number stated that the reaction force was 49 N.

Q9.

In this question, there were problems with the force diagram and resolving correctly. Many candidates were able to find the reaction force correctly, but there were more problems in part (d).

Q10.

This question was done quite well on the whole. The major source of errors was resolving incorrectly. The fact that the friction force was given did help some candidates to work towards the correct answer.

Q11.

Part (a) was usually done very well. However there were very few good responses to part (b). The most common error was to try to find the magnitude of the resultant of P and the friction, assuming that the friction force had magnitude 58.8 N. For many candidates their understanding of the friction inequality seemed poor. In part (c), the candidates tended to either find the correct value or to simply give the answer 16 N, which was the product of the mass and the acceleration. Many candidates did part (d) very well, but a few omitted the negative sign from the acceleration. Those candidates who had difficulties usually used an incorrect acceleration.

Q12.

In part (a), there were a number of candidates who simply seemed to think that the acceleration was given by distance over time. Many of these candidates also gave the units as m s^{-1} , which further confirmed that their approach was incorrect. However their attempts at part (b) were very much better, probably helped by the printed answer. There were also a number of good solutions to part (c). The most common error in part (c) was to ignore the acceleration when calculating the friction force.

Q13.

In part (a) of this question, some candidates used an incorrect angle when trying to resolve. Some others simply assumed that the normal reaction force was equal to the weight. In part (b), those who had had difficulties resolving in part (a) had further difficulties here. However there were a reasonable number of candidates who were able to find the correct coefficient of friction. A large number of candidates did not include an inequality in their working. In part (c), a good number of candidates calculated the magnitude of the friction force correctly, but some then went on to assume that this was the only force acting.

Q14.

Again there were a good number of complete solutions to this question. Some candidates did have difficulties with parts (c) and (d). This was generally due to ignoring the friction force and simply assuming that the tension was equal to the product of the mass and the acceleration.

Q15.

Drawing the force diagram in part (a) did cause difficulties for the candidates with several wanting to include a force directed down the slope. Part (b) was done correctly by almost all of the candidates. In part (c), several candidates did not form a three term equation, either omitting the component of the weight down the slope or the friction force. Most candidates produced a good attempt at part (d).

Q16.

There were many good solutions to this question. There were a number of candidates who included additional forces on their force diagrams. Candidates should be encouraged to only show the forces acting on force diagrams. Including velocities or components as well as the forces often confuses diagrams and can cause marks to be lost. Part (b) was done very well. In part (c), the most common error was only to include one force, typically the friction force, and not to include the component of the weight that acts down the slope. Two other less common errors were to use $80ga$ instead of $80a$ and to resolve the weight incorrectly or not at all.

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Q17.

Part (a) was generally done well. Some candidates made arithmetic errors and a small number misquoted the constant acceleration equation. The force diagrams were drawn very well. Most of the candidates found it easy to calculate the magnitude of the friction force, although a small number took the reaction force as being equal to the weight. Many candidates had difficulty finding the tension in the rope.

Q18.

There were a number of good responses to this question. Some parts caused the candidates a little more difficulty than the earlier questions. In part (a), some candidates included additional forces. Parts (b) and (c) were generally well done, the only significant problem was that some candidates used the wrong angle when finding the components of the weight and so ended up with their answers to parts (b) and (c) interchanged. This caused subsequent problems in part (d). However, most candidates made a good attempt at part (d). Part (e) was more challenging. Many candidates felt that the 14 kg would just

slide down the slope. Some argued that the friction and weight would be proportional to the mass and so it would stay still. Quite a number repeated all the earlier calculations to obtain the same coefficient of friction and often concluded that the block would remain at rest.

Q19.

Part (a) was generally done very well. Very few candidates did not form two equations of motion as requested. A few had problems dealing with the friction force. Part (b) proved to be more difficult for some candidates, who dropped the negative sign and simply took the acceleration to be 1.225. Many of these candidates then just dropped a negative sign in the middle of their working. Part (c) was done well by some candidates, but again the negative acceleration caused problems for some.



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